

Task: SUP - Superbajtor

Lab 8. Available memory: 256 MB.

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Description

Professor Makary is leading work on the most high-priority Bajtocki scientific project — project "Superbajtor". Superbajtor is a being with superhuman algorithmic skills... The Professor is now trying to create Superbajtor's genetic code in laboratory conditions. Superbajtor's code will be a chain of nucleotides, similar to human DNA, but containing more types of nucleotides represented by capital letters: $A, B, \dots, Z$.

The Superbajtor chain is initially empty. The professor performs successive experiments by inserting batches of identical nucleotides into selected places in the chain.

Each such operation has the form `insert(pos, c, k)`, which means inserting k copies of nucleotide c at position pos . All nucleotides located at positions $\geq pos$ are shifted to the right. Positions are numbered from 0, so $pos = 0$ corresponds to insertion at the beginning.

Superbajtor's algorithmic abilities depend on the length of the longest contiguous fragment of the chain consisting of identical nucleotides. Therefore, after each insertion operation, Professor Makary must determine the length of such a fragment to assess the current potential of the organism.

Write a program that helps the professor with this task.

Input

The first line of input contains an integer Q — the number of insertion operations performed ($1 \leq Q \leq 500\,000$).

Each of the following Q lines contains a description of one insertion operation. The description of a single operation consists of two integers $p \geq 0, k \geq 1$ and a character c (where c is a character from the range $A, B, \dots, Z$), separated by single spaces. This corresponds to inserting k copies of nucleotide c at position p .

The sum of numbers k will not exceed 10^9 . Furthermore, if K nucleotides have been inserted in total so far, then $p \leq K$, i.e., p describes a position in the chain ($p < K$) or the position just after its end ($p = K$).

Output

For each insertion operation, print one integer in a separate line — the length of the longest contiguous fragment of the chain consisting of identical nucleotides.

Example

Input:

3
0 3 A
1 2 B
3 4 A

Output:

3
2
6

Explanation of the example: After the first insertion operation, we get the chain AAA. After the second: ABBAA. After the third: ABBAAAAAA.